

Project Ability

Universal Multi-Modal Communication & Accessibility Hub

Comprehensive Hardware & Systems Proposal

Prepared for Microprocessor Engineering Development

1. Executive Summary

Project **Ability** is a highly ambitious microprocessor-based ecosystem designed to bridge the communication gap between individuals with varied sensory and speech impairments (deaf, mute, blind, and paralyzed) and the general public. Moving beyond single-purpose assistive devices, "Ability" provides a robust, multi-modal translation architecture.

By utilizing a decentralized edge-to-hub hardware topology, the system integrates wearable gesture recognition, real-time machine learning, visual avatar rendering, physical Braille actuation, health monitoring, and computer vision-based eye tracking into a single unified platform.

2. Core Objectives & Capabilities

The system is engineered to perform bi-directional translation across four primary modalities, while ensuring the safety and operational independence of the user:

- **Sign Language Translation:** Converts physical hand gestures (via a sensor-laden glove) into Text, Speech, or physical Braille output.
- **Visual Rendering:** Converts incoming Speech or Text into a 3D Sign Language Avatar displayed on a portable screen.
- **Tactile Interfacing:** Converts incoming Text into physical Braille via solenoid actuators, and allows Braille input to be converted into Text/Speech.
- **Vision-Based Navigation:** Utilizes facial landmark tracking (Eye-Tracking & Blink Detection) to allow users with severe motor impairments to control the system entirely via gaze.
- **Continuous Health Monitoring:** A decoupled, wrist-mounted hub tracks SpO2, heart rate, and utilizes inertial sensors for automated fall detection and emergency alerts.

3. System Architecture

The Edge-to-Hub Paradigm

To balance the conflicting needs of continuous wearability (health sensors) and high-performance computing (machine learning), the hardware is split into two distinct tiers: an ultra-low-power ESP32-S3 wearable edge, and a high-performance Raspberry Pi 5 central processing hub.

3.1 The Wearable Edge (ESP32-S3)

The wearable tier is designed around a continuous-wear wrist hub and a detachable sensory glove. The ESP32-S3 handles real-time analog data acquisition, runs local fall-detection algorithms on its secondary core, and streams formatted data packets via wireless protocols (BLE/Wi-Fi) to the main hub.

3.2 The Central Processing Hub (Raspberry Pi 5)

Acting as the core computational brain, the Raspberry Pi 5 (8GB) ingests edge data, executes the ML models for gesture translation, runs NLP for Speech-to-Text (STT) and Text-to-Speech (TTS), dynamically renders the 3D Sign Language Avatar, and manages the MediaPipe computer vision pipeline for eye-tracking.

4. Comprehensive Bill of Materials (BOM)

4.1 Wearable Glove & Health Hub (ESP32-S3 Edge)

Component	Specification / Purpose
Microcontroller	ESP32-S3 Development Board (WROOM) - Dual core, highly efficient BLE/Wi-Fi.
Flexion Sensors	5x Spectra Symbol Flex Sensors (2.2") + ADS1115 16-bit ADC for high-precision, low-noise analog reading.
Spatial Orientation	BNO085 9-DOF IMU (Hand orientation, eliminates yaw drift via magnetic fusion).
Health Sensors	MAX30102 (SpO2 & Heart Rate) + MPU6050 (Wrist-mounted Fall Detection).
Power & Wiring	3.7V 1000mAh LiPo + TP4056; 28 AWG Silicone-jacketed stranded wire for durability.

4.2 Vision & Eye-Tracking Subsystem

Component	Specification / Purpose
Camera Module	Raspberry Pi Camera Module 3 NoIR (Wide) - Tracks irises and facial landmarks without reliance on ambient visible light.
Illumination	850nm IR LED Module Ring - Provides invisible, high-contrast illumination for accurate computer vision processing.

4.3 Central Processing Hub (Raspberry Pi 5)

Component	Specification / Purpose
Core Computer	Raspberry Pi 5 (8GB RAM) with Active Cooler.
Storage	M.2 NVMe HAT + 256GB NVMe SSD (Required for fast model loading and UI rendering).

Audio Interface	INMP441 I2S Omnidirectional Mic (Input) + MAX98357A I2S Mono Amp & 3W Speaker (Output).
Visual Interface	7-inch DSI Capacitive Touchscreen (Main UI and Avatar display).
System Power	27W (5V/5A) USB-C Power Delivery (PD) Power Bank.

4.4 Braille Actuation Interface

Component	Specification / Purpose
Tactile Output	6x 5V Push-Pull Mini Solenoids (Arranged in a standard 3x2 Braille cell).
Driver & Protection	ULN2003 Darlington Transistor Array + 6x 1N4007 Flyback Diodes.
Tactile Input	6x Tactile Push Buttons (Perkins layout) + Spacebar.

5. Software & Algorithmic Pipeline

The software stack relies on decentralized processing to prevent bottlenecks. The **ESP32-S3** will handle immediate sensor filtering (e.g., Kalman filtering for the IMUs and ADCs). The **Raspberry Pi 5** will run a Python-based pipeline utilizing:

- **Machine Learning:** TensorFlow Lite or PyTorch for classifying the multivariate time-series data from the glove into discrete sign language gestures.
- **Computer Vision:** MediaPipe Face Mesh for real-time extraction of 478 3D facial landmarks, specifically targeting the iris and eyelid indices for Gaze Mapping and Eye Aspect Ratio (EAR) blink detection.
- **Audio Processing:** Offline speech models to ensure continuous operation independent of internet connectivity.

6. Conclusion

Project **Ability** represents a comprehensive approach to assistive technology. By distributing the computational load across specialized microprocessors and embracing a modular architecture, the

system achieves the necessary balance of performance, wearability, and multi-modal accessibility required to empower users with severe sensory and physical disabilities.